

Does a Higher Education Quality Policy Improve Outcomes? Evidence from Teacher Training Programmes

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Motivation

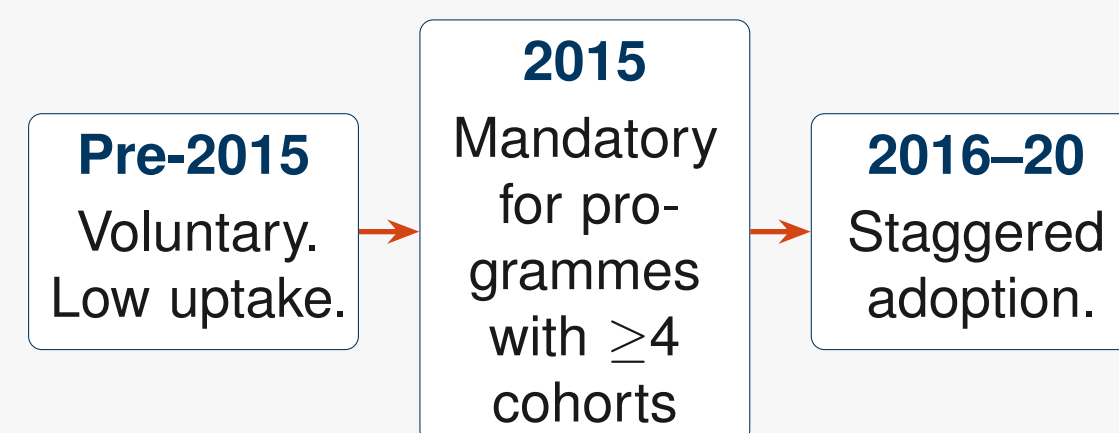
- Higher education is central to human capital formation, yet ensuring programme-level quality remains a key challenge in middle-income countries
- A growing literature finds school accountability improves student outcomes (Mansfield & Slichter, 2026; Eyles & Machin, 2019), however, evidence on higher education is scarce and mostly descriptive
- In Colombia, prior studies cannot isolate causal effects: voluntary take-up means better programmes self-select in before any policy change (Farieta, 2024)
- Colombia's 2015 reform created mandatory accreditation for teacher training, a rare instance of binding policy with exogenous variation and staggered rollout

Research Question

Does mandatory accreditation improve student learning outcomes in teacher training programmes?

Institutional Context

Colombia's 2015 Reform — Law 1753 / Decree 2450



Non-compliance ⇒ enrolment suspension or closure.

Data

Linked administrative dataset: SNIES × ICFES

Source	Content
Saber 11	Secondary exit exam; baseline ability
SNIES	Enrolment, programme status, accreditation timelines, programmes characteristics
Saber Pro	HE exit exam — main outcome

- $N = 182,990$ students; bachelor's programmes in education
- Pre-policy balance:** early/late adopters indistinguishable in Saber 11, SES, gender, size

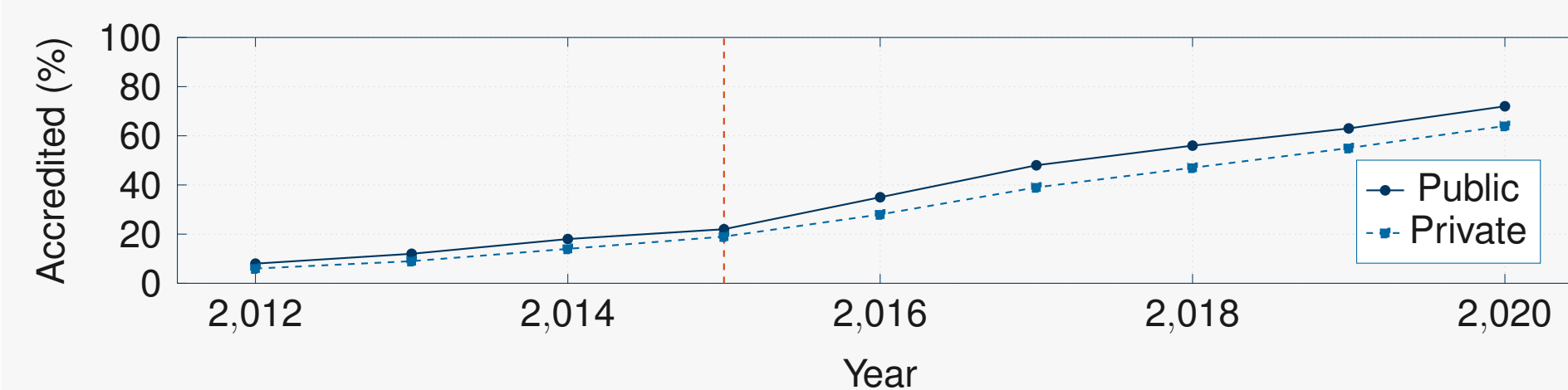


Figure 1. Share of education programmes holding high-quality accreditation. Adoption is staggered and earlier among public institutions.

Empirical Strategy

Main Design: Staggered DiD — Callaway & Sant'Anna (2021)

Adoption was *gradual* (2016–2020). Standard TWFE yields biased estimates under treatment-effect heterogeneity (CS, 2021); Therefore, I use group–time ATTs.

$$ATT(g, t) = \mathbb{E}[Y_{i,t}(1) - Y_{i,t}(0) | G_i = g]$$

- Group g : programmes first accredited in year g
- Counterfactual: *not-yet-treated* units
- Assumption:* conditional parallel pre-trends

Complementary Design: IV — Regulatory Threshold

Programmes with ≥ 4 cohorts by 2015 faced *mandatory* compliance. It is motivated by a jump in $\Pr(\text{accredited})$ from 22% to 65%.

$$\text{Acc}_{p,t} = \pi_0 + \pi_1 \cdot \text{Elig}_p + \nu_{p,t}$$

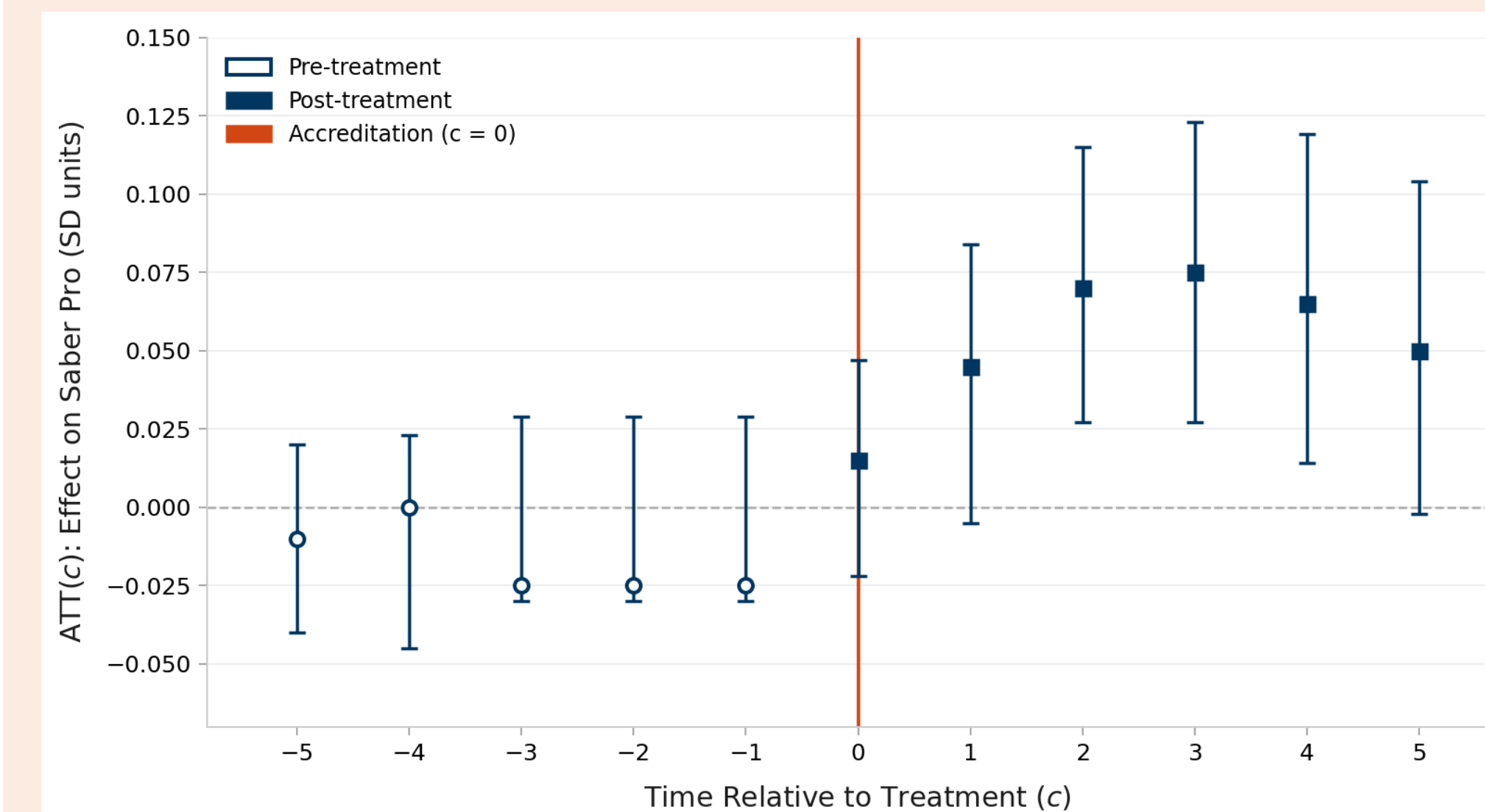
$$Y_{i,t} = \alpha + \beta \widehat{\text{Acc}}_{p,t} + \gamma X_i + \delta_t + \mu_h + \varepsilon_{i,t}$$

$$\beta = \text{LATE for threshold-compliers}$$

Why two designs?

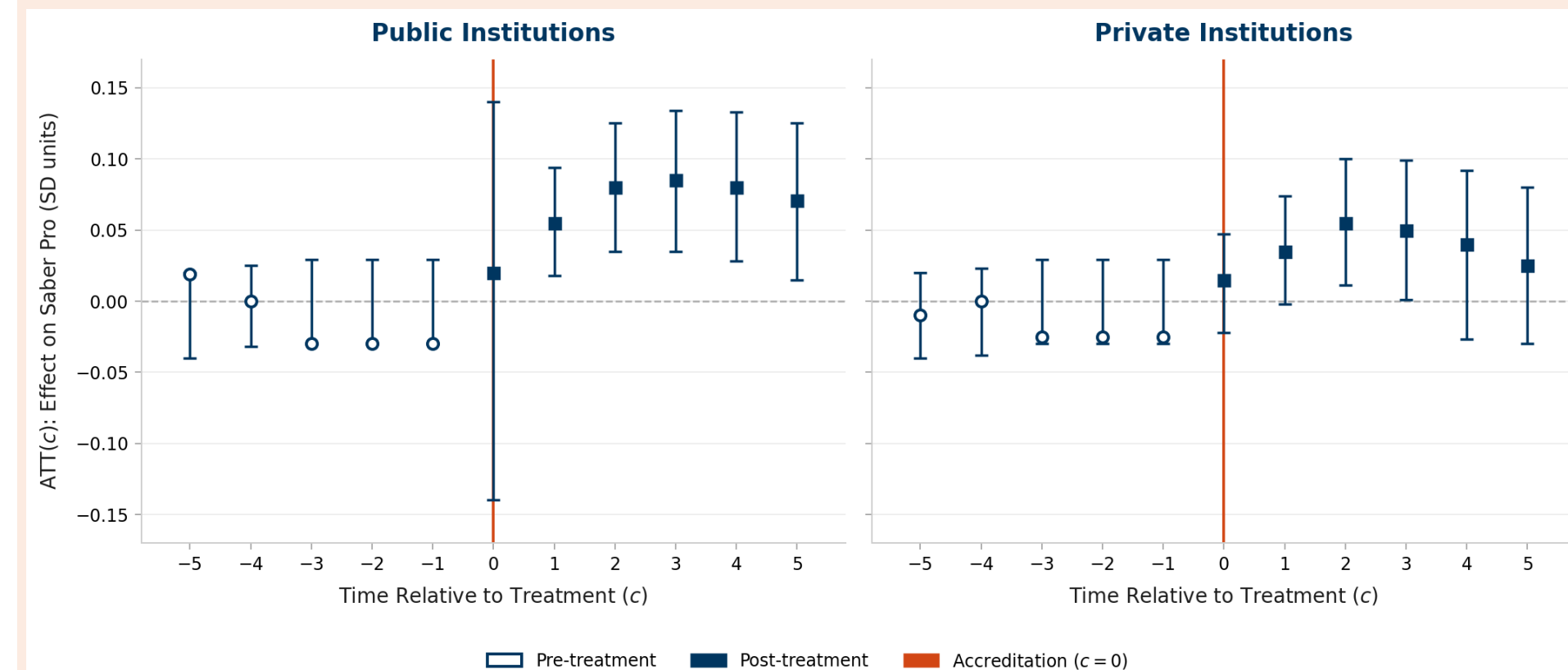
DiD exploits *timing* variation · IV exploits *eligibility* variation. Consistency strengthens causal credibility.

Main Results: Staggered DiD — Dynamic ATT



- Pre-treatment** ($c = -5$ to -1): flat and near zero — parallel trends supported
- Post-treatment:** positive from $c = 1$, reaching 0.08–0.12 SD by $c = 3$ –4
- Gradual build-up consistent with curriculum and faculty reforms

Heterogeneity by Sector — CS(2021)



- Public:** stronger, more persistent effects (≈ 0.08 – 0.09 SD at $c = 3$), consistent with greater institutional capacity for structural compliance.
- Private:** smaller gains, earlier peak, and faster fade-out, consistent with less structural compliance.

IV Result: Regulatory Threshold

	Standardised Saber Pro Test Scores			
	OLS (1)	IV (2)	OLS (3)	IV (4)
Accredited Program	0.192*** (0.027)	0.128*** (0.026)	0.179*** (0.027)	0.120*** (0.029)
Saber 11 score			0.595*** (0.004)	0.605*** (0.004)
Controls	No	No	Yes	Yes
HEI FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	182,990	182,990	182,990	182,990
First stage (Compliance)		0.945*** (0.003)		0.940*** (0.003)
F statistic		74.6		79.8

Notes: Standard errors clustered at the programme level. The dependent variable is the standardised Saber Pro test score. Columns (1)–(2) present OLS and IV estimates without additional control variables, while columns (3)–(4) introduce controls for student characteristics. *** Significant at 1%.

Conclusions & Contributions

- 0.10–0.12 SD gain in Saber Pro, comparable to school accountability reforms (Mansfield & Slichter, 2026)
- Effects gradual; larger and more persistent in public HEIs
- First causal evidence on a binding quality policy in HE via a national regulatory discontinuity

Policy implication

Binding QA can strengthen human capital formation in middle-income countries. Results suggest effects are not automatic, complementary institutional support, especially in private HEIs, may be needed.